



Collecting and analysing monitoring data: event logs and network traffic capture

420-1C1-DW Preventative Monitoring · Class 13 · Thursday, October 8, 2026 · Topic 4, Lessons 4.1 and 4.2 · AEC LEA.D8, Dawson College · Asen Grozdanov, Cyber.SoHo

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Plan of the hour

Minutes	Part	What we do
0–3	Opening	Where we are, outcomes, the Monday in raw data
3–9	4.1, Parts 1 and 2	What a log entry is, the six questions, time; the Windows channels, events, logon types, failure reasons
9–17	4.1, Part 2 continued	Audit policy; reading and exporting with Event Viewer, wevtutil and Get-WinEvent
17–22	4.1, Part 3	Syslog, rsyslog files, the journal, reading commands
22–26	4.1, Part 4	Central collection, the common schema, the seven checks, logs as personal information
26–31	Real-life example 4.1	Two Windows exports and one Linux log, codes 4.1.A and 4.1.B, Figure 4.1.2
31–37	4.2, Parts 1 and 2	Packets, frames, headers; capture points; tools, pcap, filters, snapshot length, bounded files
37–42	4.2, Parts 2 and 3	Capture record and legal frame; reading a capture: five questions
42–50	Real-life example 4.2	Capture plan and capture record, the sample capture, codes 4.2.A and 4.2.B, Figure 4.2.2
50–55	Lab 3 summary	What Lab 3 asks, what today's deliverables become
55–60	Review and next steps	What to remember, terms, Quiz 3, Class 14

Where we are and what you will be able to do

- Topic 3 ended with the alerts (Lesson 3.5, Lab 2); Topic 4 is about the data behind them
- An alert is the tool's reading of a few fields; the entries and the traffic are the evidence
- Outcome 1: read a Windows event and a syslog line field by field: when, where, source, what, who, from where, result
- Outcome 2: recognize the event numbers, logon types and failure reasons behind the Class 12 alerts; find the same on Linux
- Outcome 3: explain central collection and check a log against seven conditions
- Outcome 4: describe a frame, choose a capture point, write a capture filter, bound a capture, write a capture record
- Outcome 5: read a capture through its summaries: conversations, bytes, timing, services, handshakes, names
- Competency EZ01, element 4: accurate interpretation of the available data

The running example: the same Monday, seen in the raw data

- Ébénisterie Val-des-Pins Inc. (Laval, 22 employees), Wazuh pilot on MON-01 (Lab 2), the test Monday of Class 12
- 06:42:15 PC-ACCT-01, event 4624 type 2: acct.clerk at the keyboard before 07:00 (AL-03)
- 07:49:04 MON-01, sshd: Failed password for invalid user admin from 10.20.1.104 port 51522
- 07:52:01 to 07:53:12 SRV-FILE-01: five events 4625, admin.it from 10.20.1.104, SubStatus 0xC000006A (AL-01)
- 07:54:30 SRV-FILE-01: 4624 then 4672 for admin.it (AL-02); 07:55:50 4720 and 07:56:05 4732: svc.backup2 (AL-04)
- 07:58:40 SRV-FILE-01: 1102, the Security log is cleared by admin.it (AL-05)
- Tuesday 01:59:58 PC-ACCT-01: DNS upload.example.net = 203.0.113.50; 02:00:00 TCP 49731 to 443, 135 kB/s = 486 MB/h (AL-07)
- Addresses: Class 6 plan (10.20.1.0/24 office, FW-01 10.20.0.1); lab replica 192.168.56.0/24; outside: RFC 5737 blocks

4.1 What an event log is

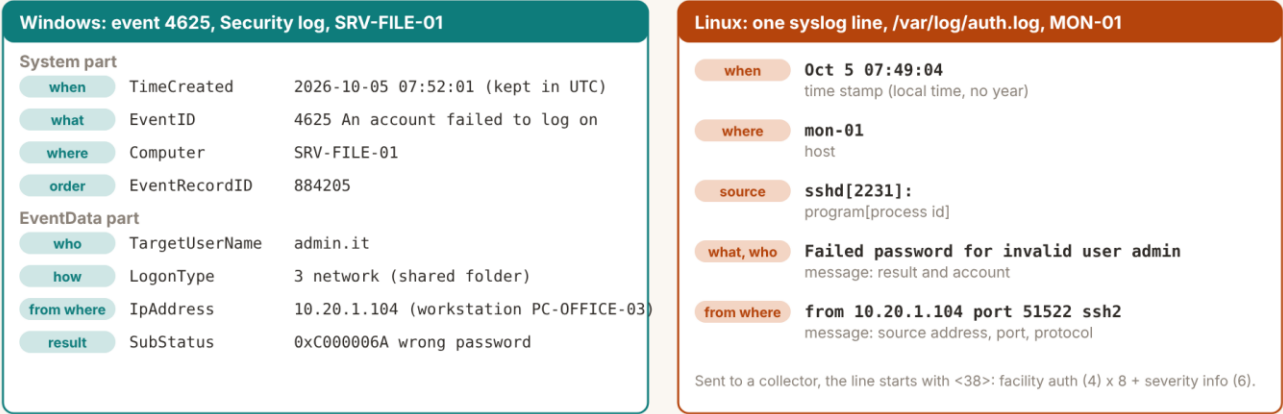
- Event: one recorded occurrence (a logon, a failed logon, a service start, a refused packet, a file created)
- Log entry: the written record of the event, made of fields (time, system, program, account, result)
- Event log: the ordered collection of entries of one source
- A log records what the system was configured to record, nothing more
- A log is written by the system it describes: whoever controls the system can clear or alter it
- Those two facts explain the audit policy (Part 2) and central collection (Part 4)

The six questions a log entry answers

- When: the time stamp, with its time zone
- Where: the system that recorded the event (host)
- Which source: the program or channel that wrote it (sshd, Security, FW-01)
- What: the event (number 4625; 'Failed password for')
- Who: the account concerned
- From where: the address or the workstation of the origin
- With what result: success or failure, and why (SubStatus 0xC000006A, 'invalid user')
- The questions are the same for every source; only the field names and the format differ (Figure 4.1.1)

Figure 4.1.1 Two log entries, the same six questions

Windows Security event 4625 of SRV-FILE-01 (left) and one line of /var/log/auth.log on MON-01 (right), Monday 07:52 and 07:49



Linux: one syslog line, /var/log/auth.log, MON-01

when

Oct 5 07:49:04
time stamp (local time, no year)

where

mon-01
host

source

sshd[2231]:
program[process id]

what, who

Failed password for invalid user admin
message: result and account

from where

from 10.20.1.104 port 51522 ssh2
message: source address, port, protocol

Sent to a collector, the line starts with <38>: facility auth (4) x 8 + severity info (6).

The common schema (Table 4.1.G): the same fields for every source

time	host	source	event	outcome	account	source address	detail
07:52:01	SRV-FILE-01	Security	4625 failed logon	wrong password	admin.it	10.20.1.104	type 3, PC-OFFICE-03
07:49:04	mon-01	sshd[2231]	Failed password	invalid user	admin	10.20.1.104:51522	ssh2

Both entries answer the same questions; only the field names and the format differ. A parser that fills the common schema makes them comparable.

Fictional data. Cyber.SoHo / Asen Grozdanov, 420-1C1-DW Preventative Monitoring, Class 13.

Figure 4.1.1.1: a Windows event 4625 and a syslog line, the same six questions



Time: the field that makes two logs comparable

- Two entries from two systems can be put in order only if the two clocks agree
- NTP (Network Time Protocol, RFC 5905): each system asks a time server and corrects its clock
- Class 6 found FW-01 and SRV-FILE-01 four minutes apart: a 07:56 firewall record and a 07:52 Windows event may be the same moment
- Lab 2: time zone set on MON-01; Lab 3: check every clock before putting events side by side
- Windows stores event times in UTC and shows local time; a syslog file writes local time; a pcap file stores UTC
- Montréal in October: EDT, UTC minus four hours
- Write the time zone once at the top of a timeline and convert everything to it

Windows event logs: the channels (Table 4.1.A)

Channel (log)	What it records	Examples of event numbers
Security	Results of the audit policy: logons and logoffs, account and group changes, use of administrative privileges, changes to the audit policy, clearing of the log	4624, 4625, 4634, 4672, 4720, 4732, 4740, 1102
System	The operating system and its services: a service installed, started or stopped, a driver loaded, a time change, a shutdown, the log service itself	7045, 7036, 6005, 6006, 1074, 104
Application	Messages written by programs: errors, warnings and information from backup software, antivirus, databases	1000, vendor-specific numbers
Microsoft-Windows-Sysmon/Operational	Present only when Sysmon is installed: process creation with its command line, network connections made by processes, file creation	1, 3, 11
Microsoft-Windows-Windows Defender/Operational	Detections, scans and definition updates of the built-in antivirus	1116, 1117, 2000
Microsoft-Windows-TaskScheduler/Operational	Scheduled tasks registered, started and completed (disabled by default on workstations)	106, 200

The structure of a Windows event

- System part, identical for every event: provider, EventID (the event number), level, TimeCreated, EventRecordID, Computer
- EventRecordID increases by one with every event written: a jump reveals a gap
- EventData part, specific to the event number: 4625 carries TargetUserName, LogonType, IpAddress, WorkstationName, Status, SubStatus
- 4732 carries the group (TargetUserName), the member (MemberSid, sometimes MemberName) and the actor (SubjectUserName)
- Event Viewer, Details tab: Friendly View or XML (Extensible Markup Language)
- The XML names are the names used by filters, scripts and Wazuh rules
- Computer is the host that wrote the event; the origin of a network logon is in IpAddress and WorkstationName

Security events to know (1/2) (Table 4.1.B)

Event	Meaning	Fields to read	Class 12 alert that uses it
4624	An account was successfully logged on	TargetUserName, LogonType, IpAddress, WorkstationName, LogonId	AL-02, AL-03
4625	An account failed to log on	TargetUserName, LogonType, IpAddress, WorkstationName, Status, SubStatus	AL-01
4634	An account was logged off	TargetUserName, LogonId (pairs with the 4624 of the same session), LogonType	Session length
4647	User initiated logoff	TargetUserName, LogonId	Session length
4648	A logon was attempted using explicit credentials (for example runas)	SubjectUserName, TargetUserName, TargetServerName	Context of AL-01
4672	Special privileges assigned to new logon (an administrative session)	SubjectUserName, PrivilegeList	Context of AL-02, AL-04
4688	A new process has been created (command line only if the policy enables it)	NewProcessName, CommandLine, ParentProcessName, SubjectUserName	Topic 5 diagnosis

Security events to know (2/2) (Table 4.1.B)

Event	Meaning	Fields to read	Class 12 alert that uses it
4720	A user account was created	TargetUserName (the new account), SubjectUserName (who created it)	AL-04
4726	A user account was deleted	TargetUserName, SubjectUserName	Account management review
4732	A member was added to a security-enabled local group	MemberSid or MemberName, TargetUserName (the group), SubjectUserName	AL-04
4740	A user account was locked out	TargetUserName, TargetDomainName (the caller computer)	Follows a burst of 4625
4719	System audit policy was changed	SubjectUserName, CategoryId, SubcategoryGuid, AuditPolicyChanges	Loss of visibility
1102	The audit log was cleared	SubjectUserName	AL-05

Logon types: how the account reached the computer (Table 4.1.C)

Logon type	Name	How the account reached the computer	Normal example at Val-des-Pins
2	Interactive	At the keyboard of the computer	acct.clerk opening a session on PC-ACCT-01
3	Network	From another computer, through a shared folder, a printer or a remote administration tool	PC-ACCT-01 opening a shared folder of SRV-FILE-01
4	Batch	A scheduled task started by the task scheduler	The nightly export of the accounting program
5	Service	A service started by the service control manager	svc.backup at 07:17 and 08:17
7	Unlock	The screen was unlocked	Return from a break
8	NetworkCleartext	Network logon where the password travelled in clear text	Should not happen
9	NewCredentials	The same session continues with other credentials (runas /netonly)	Rare on a workstation
10	RemoteInteractive	Remote desktop	The external accountant's remote session, on request
11	CachedInteractive	At the keyboard, with cached domain credentials, without a domain controller	A laptop away from the office

Failure reasons: the SubStatus field of event 4625 (Table 4.1.D)

SubStatus code	Meaning	What it tells the technician
0xC000006A	The password is wrong	A person or a program trying passwords; a typo when it happens once
0xC0000064	The account does not exist	Someone is trying account names, not only passwords
0xC0000234	The account is locked out	The lockout policy worked; the burst happened before
0xC0000072	The account is disabled	An old account is being used; who knows its name?
0xC000006F	Logon outside the allowed hours	The account has time restrictions; the attempt was outside them
0xC0000070	Logon from a workstation that is not allowed	The account has workstation restrictions
0xC0000071	The password has expired	Usually an expected failure after a password policy change
0xC0000193	The account has expired	A temporary account (contractor, intern) still in use

What is recorded depends on the audit policy

- A Security event is written only if its audit setting is enabled: success, failure or both
- Workstations record logon success and failure by default; servers record more
- 4688 (process created) needs 'Audit Process Creation'; its command line needs 'Include command line in process creation events'
- Microsoft publishes recommended settings (audit policy recommendations, links section)
- Read the active policy: `auditpol /get /category:*`
- Read the size and retention of a channel: `wevtutil gl Security`
- A workstation keeps a limited number of megabytes and overwrites the oldest events: the second reason a local log is not a complete record

Reading and exporting Windows events

- Event Viewer: Filter Current Log by event number, time, account; XML filter for precise conditions
- `wevtutil qe Security /q:"*[System[(EventID=4625)]]" /c:20 /f:text` (the last twenty failures)
- `wevtutil epl Security C:\exports\security.evtx` (export the whole channel)
- PowerShell `Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624,4625; StartTime=...}`
- `ToXml()` gives every `EventData` field by name; `Export-Csv` writes one event per line
- The export of Real-life example 4.1 has twelve columns: `TimeCreated`, `EventID`, `Computer`, `RecordId`, `TargetUserName`, `LogonType`, `IpAddress`, `WorkstationName`, `SubStatus`, `SubjectUserName`, `MemberName`, `Detail`
- In 4732 the member is often a SID (security identifier, S-1-5-21-...): resolve the name

Linux logs: the syslog standard (RFC 5424)

- Every message carries a facility (where it comes from) and a severity (0 Emergency to 7 Debug, Table 3.5.I of Class 12)
- Priority = facility x 8 + severity: <38> is auth (4 x 8 = 32) plus informational (6)
- Facilities (Table 4.1.F): kern, user, mail, daemon, auth, syslog, cron, authpriv, ntp, local0 to local7 (FW-01, SW-01)
- A line in a file: time stamp, host name, program[process number], message
- Traditional time stamp 'Oct 5 07:49:04': no year, no zone; ISO 8601 '2026-10-05T07:49:04.123456-04:00': both
- The message is free text written by the program: know the formats of sshd, sudo, PAM (Pluggable Authentication Modules)
- rsyslog receives the messages and writes the files of /var/log on Ubuntu Server 24.04 (MON-01)

Where the Linux logs are (Table 4.1.E)

File or store	What it contains	How to read it
/var/log/auth.log	SSH (Secure Shell) logins and failures, sudo commands, PAM sessions, cron sessions, user and group changes	<code>sudo grep sshd /var/log/auth.log; sudo tail -f /var/log/auth.log</code>
/var/log/syslog	Everything the system and the daemons write, including the Wazuh manager messages	<code>sudo grep wazuh /var/log/syslog</code>
/var/log/kern.log	Kernel messages: network interfaces up and down, disk errors, firewall (ufw) decisions	<code>sudo grep -i ufw /var/log/kern.log</code>
/var/log/ufw.log	Blocked and allowed packets of the host firewall, when logging is on	<code>sudo tail /var/log/ufw.log</code>
/var/log/dpkg.log and /var/log/apt/history.log	Packages installed, upgraded and removed (the update log of Lab 2, Criterion 3.3)	<code>grep upgrade /var/log/apt/history.log</code>
/var/log/wtmp, /var/log/btmp, /var/log/lastlog	Sessions opened (wtmp), failed logins (btmp), last login per user (lastlog)	<code>last; sudo lastb; lastlog</code>
/var/log/journal/ (the journal)	Every message of every service, with structured fields, from boot to boot	<code>journalctl -u ssh --since "2026-10-05 07:00"</code>
/var/ossec/logs/alerts/alerts.json and archives/archives.json	The alerts (always) and every event received (when logall_json is enabled): the central copy of the other hosts' logs	<code>sudo tail -n 200 /var/ossec/logs/alerts/alerts.json</code>
/var/log/*.1, *.2.gz	The rotated and compressed older copies (weekly rotation, a limited number kept)	<code>zgrep sshd /var/log/auth.log.2.gz</code>

The journal, journalctl and log rotation

- systemd-journald keeps a binary, indexed journal with structured fields (unit, process, priority, boot)
- `journalctl -u ssh --since "2026-10-05 07:00" --until "2026-10-05 09:00"`
- `journalctl -p err -b` (errors since the last boot); `journalctl -o json` (one JSON object per entry)
- On Ubuntu 24.04 the journal receives everything first and rsyslog writes the files from it
- logrotate renames the current file to .1 every week, compresses the older ones (.2.gz) and deletes the oldest
- Local retention is short on both systems: central retention is the one that counts

Reading Linux logs: four commands

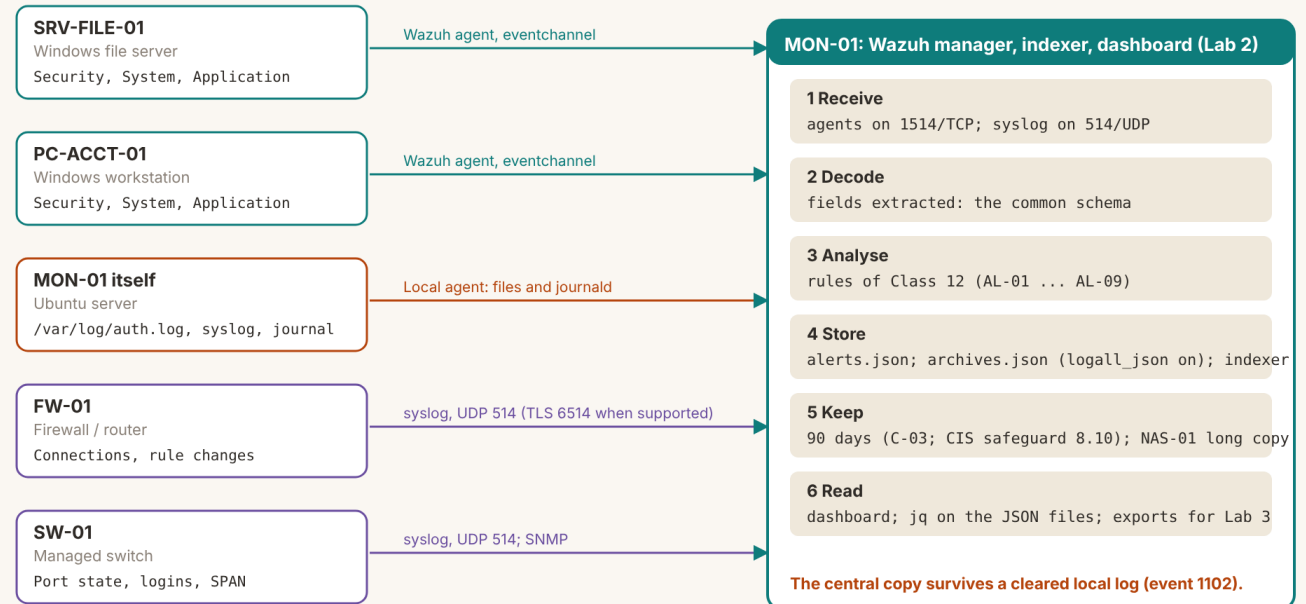
- `grep` keeps the lines that match: `sudo grep "Failed password" /var/log/auth.log`
- `awk` extracts fields from each line and prints them in the order you choose
- `sort` and `uniq -c` count the repeats: a list of source addresses becomes a table of failures per address
- `tail -f` follows a file while it grows: how you watch a Lab 3 test as it happens
- Locate fields by the text around them (after 'from', after 'port'), not by their column position: time stamp formats change
- The same method on both systems: filter the entries, extract the fields of the common schema, count, order by time
- Code example 4.1.B does exactly this on the Monday log of MON-01

Why the logs are collected centrally (Figure 4.1.3)

- 07:58:40: the Security log of SRV-FILE-01 is cleared; its thirteen morning events survive only because the agent had already sent them to MON-01
- Central collection: copy every entry, as it is written, to a system the monitored hosts cannot modify
- Three purposes: survive clearing, filling and failure; search all hosts together in one format; decide the retention in the plan
- Retention 90 days (constraint C-03, CIS safeguard 8.10); NIST SP 800-92 and the CCCS guidance describe the practice

Figure 4.1.3 From each source to the central copy on MON-01

Collection paths of need N-01 and of activity A1 (Class 6), as configured in Lab 2



Not collected: CNC-PC-01 (vendor contract, no agent; only the FW-01 records), the guest Wi-Fi (excluded, C-05), PRN-01 (no log).
Fictional network. Cyber.SoHo / Asen Grozdanov, 420-1C1-DW Preventative Monitoring, Class 13.

Figure 4.1.3: the collection paths of Val-des-Pins as configured in Lab 2

The common schema (Table 4.1.G)

Field	Windows Security event	Linux syslog line	Capture (Lesson 4.2)
time	TimeCreated (stored in UTC, shown in local time)	The time stamp at the start of the line (local time; no year in the traditional format)	Packet time stamp (UTC in the file)
host	Computer	The host name after the time stamp	The capturing interface (MON-01)
source	Channel and Provider (Microsoft-Windows-Security-Auditing)	program[pid] (sshd[2231], sudo, CRON)	Protocol decoded (DNS, TCP, TLS)
event	EventID (4625)	The message text (Failed password for ...)	Flags and message types (SYN, query, ClientHello)
outcome	4624 or 4625; SubStatus	Accepted or Failed; invalid user	Handshake completed, reset, no answer
account	TargetUserName, SubjectUserName	The user name in the message	Not visible (addresses only)
source address	IpAddress, WorkstationName	from 10.20.1.104 port 51522	Source address and port
detail	LogonType, PrivilegeList, MemberName	Method (publickey, password), command (sudo)	Bytes, DNS name, TLS server name

Seven checks before trusting a log (Table 4.1.H)

Check	Question	What goes wrong otherwise
1. Clock	Is the clock of the source synchronized (NTP) and is the time zone known?	Events from two systems cannot be put in order
2. Coverage	Does the audit policy record what the monitoring need requires?	The event never existed: nothing to find
3. Completeness	Are there gaps: cleared logs, a full log that overwrote old events, a silent source?	A gap read as 'nothing happened'
4. Integrity	Can the log be modified by the people it describes?	Evidence lost or altered
5. Retention	How long are the events kept, locally and centrally?	The event is gone when the question is asked
6. Access	Who can read the log?	Personal information read by people who have no need
7. Format	Is the format known and stable (time stamp format, field names, version of the tool)?	A parsing script silently reads the wrong field

Logs are personal information

- A log entry names a person's account, the time of their work and the address of their computer
- Québec: Act respecting the protection of personal information in the private sector, as amended by Law 25; federally: PIPEDA
- Collect the minimum, protect it, give access only to the people who need it, delete it when no longer needed
- Legal Notice section 7.2: every log, baseline and alert record of the lab is personal information, stays in the lab, is deleted after marking
- No collector or agent on any device that is not yours or a Designated Target (sections 6.3 and 7.2)
- Monitoring a person (partner, relative, roommate, classmate, colleague) is never a course activity

Links: [Legal Notice, section 7.2](#)



Videos for Lesson 4.1 (under 15 minutes each)

- Professor Messer, Log Data (CompTIA Security+ SY0-701, 4.9), 13:41: firewall logs, the Windows Event Viewer, /var/log, endpoint logs, central collection
- Professor Messer, Log Management (CompTIA Security+ SY0-601, 4.3), 9:51: syslog facilities and severities, rsyslog, journalctl, flow records (NetFlow, sFlow, IPFIX)
- Links checked as live on October 8, 2026; durations from the creator's course index
- External videos: no affiliation; their own terms apply (Legal Notice sections 5, 14, 15)
- TikTok: no direct links in this course; search #eventviewer, #syslog and apply the four-question check of Class 12

Links: [Log Data \(YouTube\)](#) [Log Management \(YouTube\)](#)



Real-life example 4.1: the raw entries behind the Monday alerts

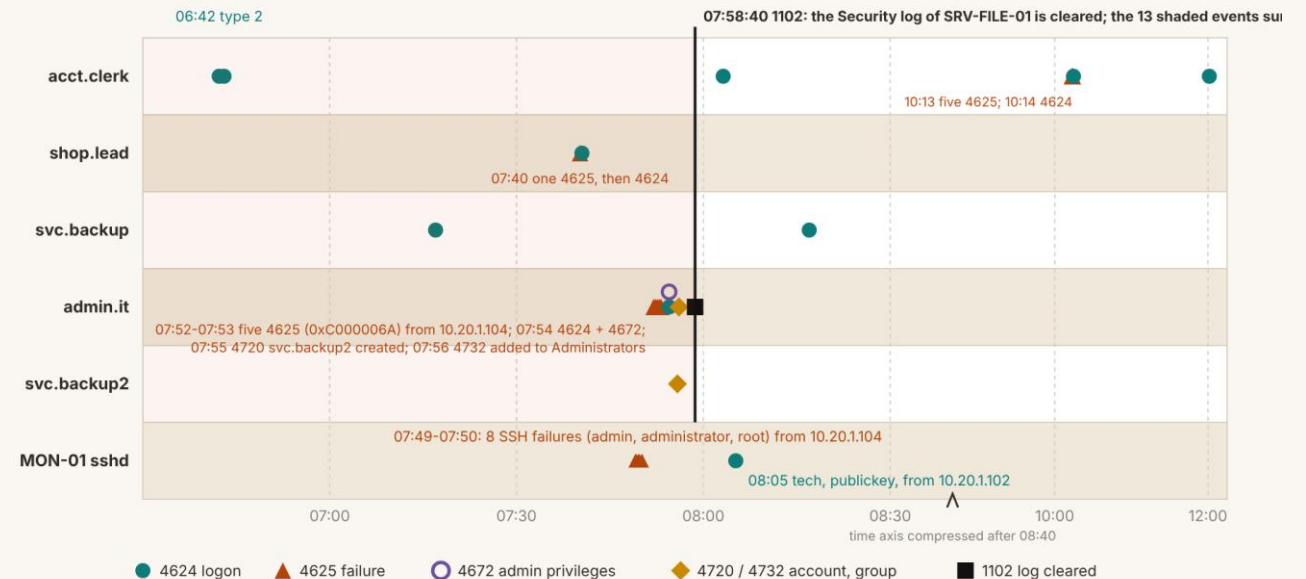
- Task: collect the entries behind the Class 12 alerts from two operating systems, decode the fields, merge them into one evidence timeline with per-account and per-source summaries, and note what the cleared log did
- Link with the lecture: the six questions, Tables 4.1.B to 4.1.D, the syslog format and Table 4.1.E, the common schema (4.1.G), the seven checks (4.1.H)
- Three places: the local export of SRV-FILE-01 (9 events, starts at 07:58:40 with 1102), the central copy on MON-01 (24 events of both Windows hosts), the authentication log of MON-01 (38 lines)
- Alerts are not evidence; they are the tool's reading of the evidence

What the two logs say, read together (Figure 4.1.2)

- Windows: five 4625 for admin.it from 10.20.1.104 in 71 s, all 'wrong password'; 4624 and 4672 at 07:54:30; 4720 and 4732 (svc.backup2); 1102 at 07:58:40
- Linux: eight SSH failures from the same 10.20.1.104 between 07:49:04 and 07:50:10 for admin, administrator and root; then the technician's own login at 08:05
- The programs state the facts; the diagnosis (which workstation, which person) belongs to Lab 3 and Topic 5

Figure 4.1.2 The Monday morning in the raw events (Windows 4624, 4625, 4672, 4720, 4732, 1102 and Linux sshd)

Output of codes 4.1.A and 4.1.B: 24 Windows events of two hosts merged with the SSH activity of MON-01 (fictional data)



Fictional data of Ebenisterie Val-des-Pins. Cyber.SoHo / Asen Grozdanov, 420-1C1-DW Preventative Monitoring, Class 13.

Figure 4.1.2: the Monday morning in the raw events, one lane per account

Code example 4.1.A: the evidence timeline (Python)

- Reads the central copy (24 events) and the local export (9 events); decodes EventID, LogonType and SubStatus with three dictionaries
- Normalizes each event into the common schema (time, host, source, event, outcome, account, source address, detail, actor)
- Merges the two copies without duplicates: key = host, record number, time, event number
- Integrity check: the local export starts with record 1, event 1102, so 13 earlier events survive only in the central copy
- Prints the timeline (24 lines) and the per-account summary; writes class13_evidence_timeline_output.csv
- admin.it: 4625 x5, 4624 x1, 4672 x1, 4732 x1, 1102 x1; acct.clerk: 4625 x5 followed by 4624 and nothing administrative
- Run: `python3 class13_windows_events_timeline.py` (Python 3.8 or later, no extra package)

Code example 4.1.B: the authentication log of MON-01 (bash)

- Reads class13_mon_01_auth.log (38 lines) or /var/log/auth.log on your MON-01
- grep keeps the sshd 'Failed password' lines (8); awk prints time, user, source, port, known or unknown account
- The function time_of() reads the time from the traditional or the ISO 8601 time stamp
- sed, sort and uniq -c count the failures per source: 8 from 10.20.1.104, flagged (limit 3)
- Accepted logins: tech, publickey, from 10.20.1.102 at 08:05:12; sudo commands: two, to read the alerts and the archive
- Writes class13_mon_01_auth_summary_output.csv (11 rows)
- Run: bash class13_linux_auth_summary.sh or sudo bash class13_linux_auth_summary.sh /var/log/auth.log

Real-life example 4.1: deliverables and the lab they feed

- Deliverable 1: the evidence timeline (class13_evidence_timeline_output.csv, 24 rows) and the per-account summary, from Code 4.1.A
- Deliverable 2: the authentication summary of MON-01 (class13_mon_01_auth_summary_output.csv, 11 rows) with failures per source, from Code 4.1.B
- The checks applied (Table 4.1.H): NTP since Lab 2; policy covers logons, account management, privilege use; local gap until 07:58:40, central copy complete; the clearing account has no access to MON-01; 90 days; exports readable by the technician only; versions recorded
- In Lab 3 the same programs, run on your own exports, produce the evidence section of your anomaly report

Flash cards and self-check 4.1

- Five flash cards: the six questions; the six event numbers; logon types 2, 3 and 10; the four parts of a syslog line and <38>; why central collection and 90 days
- Self-check: four questions with one answer, two with several answers (select 3)
- Every option has its own explanation; Hint helps without giving the answer
- Do them tonight, before Quiz 3
- On the lecture page, under Lesson 4.1

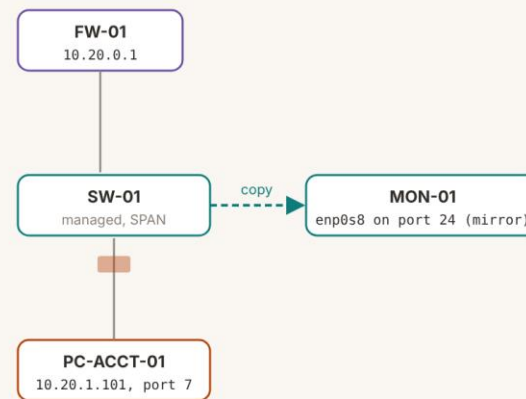
4.2 Packets, frames and headers

- A log is written by the system that acts; a capture is a copy of what travelled on the wire, made by an observer the system does not control
- Packet: a unit of limited size carrying part of the data and the information needed to deliver it; on Ethernet the unit seen is the frame (at most 1514 bytes)
- Each layer adds a header, fixed fields placed in front of the data it carries (the payload): this nesting is encapsulation
- Ethernet 14 B: MAC addresses and type; IPv4 20 B: addresses, protocol, length, TTL; TCP 20 B: ports, sequence and acknowledgement numbers, flags SYN ACK FIN RST PSH, window
- UDP 8 B: ports and length (DNS, NTP, syslog)
- Table 4.2.D lists, layer by layer, the fields that answer the monitoring questions

Figure 4.2.1 The capture point and the frame

Company network (SPAN port) and lab replica (promiscuous adapter); the layers of one frame of the night upload (not to scale)

At Val-des-Pins: SW-01 mirrors port 7 (PC-ACCT-01) to port 24 (MON-01)



The frame goes to FW-01 as usual; the switch sends a copy to MON-01. MON-01 only listens: nothing is sent back, nothing is changed.

In your lab (Lab 3): the host-only network plays the mirrored segment

VirtualBox host-only network 192.168.56.0/24
MON-01 192.168.56.10, Adapter 2: Promiscuous Mode = Allow All
PC-ACCT-01 192.168.56.21, SRV-FILE-01 192.168.56.20
Designated Targets only (Legal Notice 7.2); nothing bridged to a real network

Fictional network. Cyber.SoHo / Asen Grozdanov, 420-1C1-DW Preventative Monitoring, Class 13.

Figure 4.2.1: the capture point and one frame of the night upload, layer by layer

One frame of the upload: 1514 bytes on the wire, 256 kept (snaplen)

Ethernet 14 B
dst MAC FW-01, src MAC PC-ACCT-01, type 0x0800
IPv4 20 B
10.20.1.101 -> 203.0.113.50, proto 6, len 1500, TTL 128
TCP 20 B
49731 -> 443, seq, ack, flags ACK, window
TLS record 1460 B
type 23 application data, length 1455: encrypted
kept: 54 bytes of headers + 202 bytes of the record
not kept: the remaining 1258 bytes (snapshot length 256)
pcap record header: orig_len 1514, incl_len 256; count with orig_len

What stays visible when the content is encrypted

- addresses, ports, protocol: who, with whom, which service
- sizes and timing: how much, when, how fast
- DNS lookups: the name asked, the address answered
- the server name in the TLS ClientHello (SNI)
- the handshake: SYN, SYN-ACK, ACK; FIN or RST at the end

Not visible: file names, account, content. Those come from the logs (Lesson 4.1).

Header fields that answer the monitoring questions (Table 4.2.D)

Layer	Header	Fields that matter	Question answered
2	Ethernet (14 bytes)	Destination MAC, source MAC, EtherType (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6)	Which device on the local network sent the frame
3	IPv4 (20 bytes without options)	Source address, destination address, protocol (6 TCP, 17 UDP, 1 ICMP: Internet Control Message Protocol), total length, TTL, identification	Who talks to whom, how much
4	TCP (20 bytes without options)	Source port, destination port, sequence and acknowledgement numbers, flags (SYN, ACK, FIN, RST, PSH), window	Which service, how the connection opened and closed, how much data
4	UDP (8 bytes)	Source port, destination port, length	Which service (53 DNS, 123 NTP, 514 syslog)
7	DNS (over UDP 53)	Transaction id, flags (query or response), question name, answers (name, type, TTL, address)	Which name the client asked for, which address it received
7	TLS record and ClientHello (over TCP 443)	Record type (22 handshake, 23 application data), version, the server name extension (SNI) in the ClientHello	Which server name the client wanted; the content itself stays encrypted
7	Application data (TLS type 23)	Only the length and the timing	How much and when; not what

What encryption hides, and what stays visible

- Visible: addresses, ports and protocol (who, with whom, which service)
- Visible: sizes and timing (how much, when, how fast)
- Visible: the DNS lookup (which name was asked, which address answered, for how long)
- Visible: the server name in the TLS ClientHello (SNI, Server Name Indication)
- Visible: the handshake SYN, SYN-ACK, ACK, and the FIN or RST at the end
- Not visible: the content, the file names, the account
- The program and the account come from the host logs (event 4688, Sysmon): the two lessons meet there

Where to capture: the capture points (Table 4.2.A)

Capture point	How it works	Sees	Val-des-Pins / lab
On the host itself	The capture tool runs on the computer being watched (tcpdump, dumpcap)	Everything that enters and leaves that computer's interface	Not on PC-ACCT-01: the person using it must not be able to alter the evidence; used on MON-01 for its own traffic
Mirror port (SPAN)	The managed switch copies the frames of chosen ports to a monitoring port	All the traffic of the mirrored ports, without touching the production link	SW-01 supports it; SW-02 (workshop) does not
Network TAP	A device inserted in a link copies the signal to monitoring outputs	Every bit of the link, at full speed	Possible on the FW-01 link; not needed for this investigation
Promiscuous adapter in the lab	The VirtualBox host-only adapter of MON-01 set to Promiscuous Mode: Allow All	All the frames of the host-only network 192.168.56.0/24	The lab stand-in for the SPAN port (Lab 3)
Flow records (not a capture)	FW-01 exports one record per connection: addresses, ports, bytes, duration	Who talked to whom, how much, when	Activity A5b of Class 6, accepted; the capture is the targeted complement

In the lab: promiscuous mode on the host-only adapter

- No physical switch in the lab: the VirtualBox host-only network plays the mirrored segment
- MON-01, Settings, Network, Adapter 2 (host-only), Advanced, Promiscuous Mode: Allow All
- The interface then accepts every frame on 192.168.56.0/24, not only the frames addressed to it
- This is how Lab 3 captures the traffic of PC-ACCT-01 (192.168.56.21) on MON-01 (192.168.56.10)
- Designated Targets only; nothing bridged to a real network (Legal Notice sections 6.4 and 7.2)
- Oracle VirtualBox manual, chapter 6, documents the setting

Tools and the file format: libpcap, tcpdump, Wireshark, pcap and pcapng

- libpcap (Npcap on Windows) is the library under every capture tool
- tcpdump: the command-line tool of servers without a display (MON-01); Wireshark: the graphical analyser; dumpcap and tshark: its command-line tools
- Usual method: capture with tcpdump on MON-01, copy the file, read it in Wireshark on a computer with a display
- pcap: a 24-byte global header (magic number, version, snapshot length, link type), then one record per frame: time stamp in UTC, bytes kept, bytes on the wire
- pcapng: the newer format Wireshark saves by default (comments, several interfaces, name resolution)
- editcap -F pcap converts pcapng to pcap for tools that read only the older format (Code 4.2.B)

Capture filter versus display filter (Table 4.2.C)

Purpose	Capture filter (pcap-filter, tcpdump and Wireshark capture)	Display filter (Wireshark, after the capture)
Only one host	host 10.20.1.101	ip.addr == 10.20.1.101
Only DNS	port 53	dns
Everything except syslog to MON-01	not port 514	!(udp.port == 514)
One direction	src host 10.20.1.101 and dst port 443	ip.src == 10.20.1.101 && tcp.dstport == 443
Connection openings (SYN without ACK)	tcp[tcpflags] & (tcp-syn tcp-ack) == tcp-syn	tcp.flags.syn == 1 && tcp.flags.ack == 0
TLS server names announced by clients	(not possible: content, not header)	tls.handshake.extensions_server_name
One conversation	host 10.20.1.101 and host 203.0.113.50 and port 49731	tcp.stream eq 1

Shaping a capture: snapshot length, bounded files, no name resolution (Table 4.2.B)

Option	Meaning	In the capture plan
-i enp0s8	Interface to capture on	The mirrored interface of MON-01
-n	Do not resolve names (no DNS queries made by the capture host; faster, and the capture host stays silent)	Always
-s 256	Snapshot length: keep the first 256 bytes of each frame	Headers and the TLS server name are kept; most content is not
-w file	Write the frames to a file in pcap format instead of printing them	pc-acct-01_%Y-%m-%d_%H%M.pcap
-G 3600	Start a new file every 3600 seconds; the file name needs strftime fields	One file per hour
-W 8	With -G: stop after 8 files. With -C: keep 8 files and overwrite the oldest (rotating buffer)	The capture stops by itself after eight hours
-C 500	Start a new file every 500 megabytes (size-based rotation)	Alternative for a long capture
-Z tech	After opening the interface, continue as user tech (files owned by tech, fewer privileges)	Always
-r file	Read a file instead of an interface	tcpdump -n -r file.pcap 'port 53'
-tttt	Print the full date and time of each packet	When reading

Storage arithmetic: full capture, headers only, flow records (Table 4.2.F)

Method	Volume per second	Per day	Over 90 days	Decision
Full packet capture of the Internet link (A5)	30 Mbit/s = 3.75 MB/s	324 GB	about 29,000 GB	Rejected in Class 6: 24 times the NAS
Flow records from FW-01 (A5b)	40 records/s x 120 bytes = 4.8 kB/s	0.41 GB	37 GB	Accepted: who, with whom, how much, when
Targeted capture of one host, full frames	135 kB/s during the upload	0.49 GB per hour of upload	Bounded by the capture window	Lab 3 investigation, one night
Targeted capture of one host, snapshot 256 bytes	about 23 kB/s during the upload (256 of 1514 bytes)	0.08 GB per hour of upload	Bounded by the capture window	The plan of Real-life example 4.2

The capture record (Table 4.2.E) and the legal frame

- A capture that cannot be explained is not evidence: authorization, operator, start time (UTC and local), capture point, filter, snapshot length, tool and version, files with sizes and SHA-256 hashes, stop time, storage, deletion date
- The hash proves later that the file analysed is the file captured; Code 4.2.A writes the record
- Criminal Code section 184: knowingly intercepting a private communication is an offence; capturing other people's traffic on a shared network is interception
- Legal Notice section 7.2: capture only on systems you own or Designated Targets, for the exercise only, never in promiscuous mode on a shared network; every capture is personal information
- Val-des-Pins: the guest Wi-Fi is excluded by constraint C-05; the example capture is one host, by filter, one night, with its authorization reference
- Section 6.3 lists capturing on networks you do not control among the prohibited activities

Links: [Criminal Code, section 184](#) [Legal Notice, section 7.2](#)



Reading a capture: five questions, in this order

- 1. Who talks with whom? Conversations: two endpoints (address and port) and a protocol; Wireshark Statistics, Conversations
- 2. How much? Bytes in each direction; top talkers; a workstation that sends more than it receives is unusual
- 3. When? The time column and the bytes per second (Statistics, I/O Graphs)
- 4. Which service? Destination port, the DNS name asked just before, the server name of the ClientHello
- 5. How did it start and end? The three-way handshake (SYN, SYN-ACK, ACK) shows a normal opening by a client program; FIN closes, RST aborts, a SYN without answer failed
- Only then open individual frames; Protocol Hierarchy shows which conversation dominates

Display filters on the sample capture, and the question a capture cannot answer

- dns: the two lookups (update.example.com, upload.example.net)
- tcp.flags.syn == 1 && tcp.flags.ack == 0: the two connection openings (ports 49722 and 49731)
- tls.handshake.extensions_server_name: the two server names announced
- tcp.stream eq 1: the whole upload conversation; Statistics, Protocol Hierarchy: one conversation dominates
- Which program on the host sent the traffic? Not in the capture: event 4688 with its command line, or the Sysmon network connection event (3)
- The capture gives the destination, the volume and the time; the host log gives the process and the account; Lab 3 asks for both

Videos for Lesson 4.2 (under 15 minutes each)

- Chris Greer, Wireshark for Beginners: Capture Network Traffic, 10:05: a first capture, the interface, start, stop, save
- Chris Greer, Intro to Wireshark, Lesson 4: Where do we capture network traffic? How?, 7:30: the capture points and their trade-offs
- Professor Messer, Hardware Tools (CompTIA Network+ N10-009, 5.5), 8:40: network taps and port mirroring (optional)
- Reproduce the demonstrations only inside your isolated lab
- Links checked as live on October 8, 2026; external videos, no affiliation (Legal Notice sections 5, 14, 15)

Links: [Capture Network Traffic \(YouTube\)](#) [Where do we capture? \(YouTube\)](#) [Hardware Tools \(YouTube\)](#)



Real-life example 4.2: what left PC-ACCT-01 at 02:00?

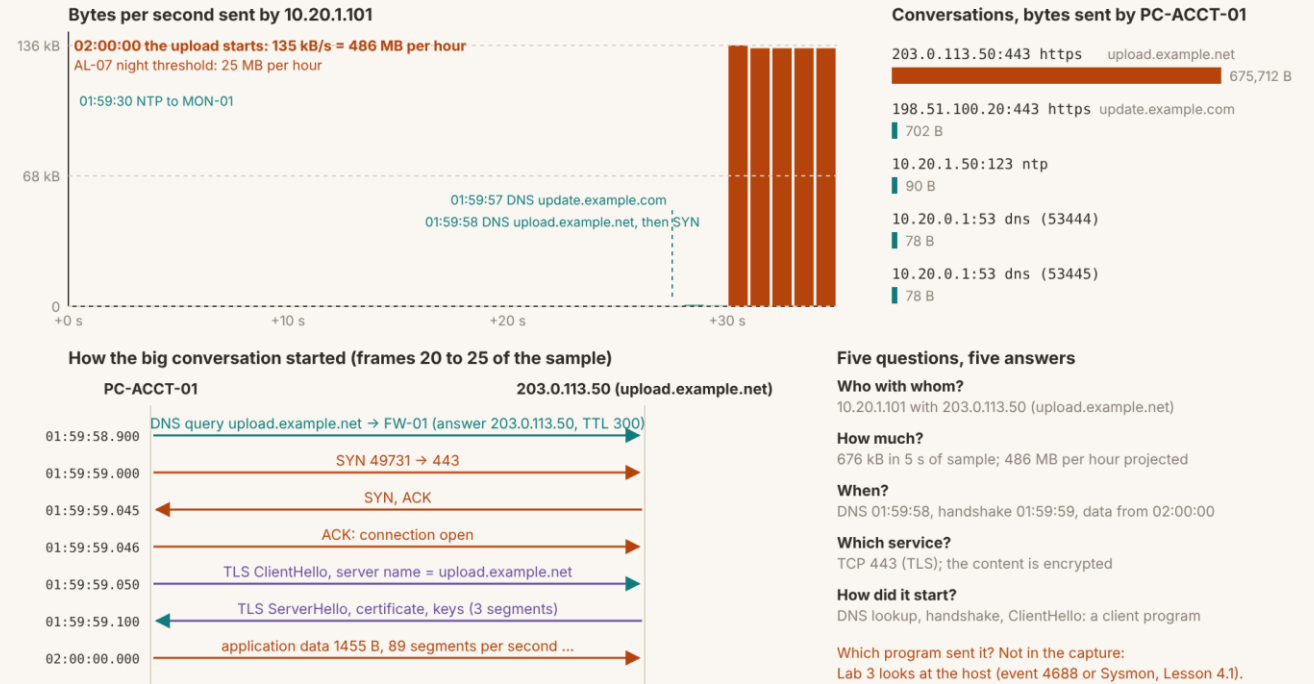
- Task: plan an authorized capture of PC-ACCT-01 for the following night, with a capture filter, a snapshot length, timed files and a capture record; then read the capture
- Link with the lecture: Tables 4.2.A to 4.2.F, Figure 4.2.1, the five reading questions
- The plan: mirror port of SW-01 (port 7 to port 24, MON-01); in the lab, the promiscuous host-only adapter
- Filter host 10.20.1.101 and not port 514; snapshot 256 bytes; one file per hour, stop after eight; /captures on MON-01, owned by tech
- Authorization: the Lab 3 guide names PC-ACCT-01 as a Designated Target; Annex A signed (a real company: the owner's written authorization, Annex D)
- Code 4.2.A checks the interface, the tool and the free space, prints the command, appends the record with the SHA-256 of the first file; nothing is captured without --run

What the capture showed (Figure 4.2.2)

- 01:59:30 NTP to MON-01; 01:59:57 DNS update.example.com = 198.51.100.20 (TTL 3600 s), a 13-frame TLS session with the vendor update server (allowlisted)
- 01:59:58 DNS upload.example.net = 203.0.113.50 (TTL 300 s); 01:59:59 handshake 49731 to 443 and ClientHello naming upload.example.net
- From 02:00:00: 1514-byte frames, 89 per second: 135 kB/s = 486 MB per hour, the AL-07 figure, against a night threshold of 25
- The capture says: the host opened the connection itself, after a lookup, to a server not on the vendor list, at a rate that fills the hour, with encrypted content; it does not say which program or account

Figure 4.2.2 What left PC-ACCT-01 at 02:00: the sample capture read by code 4.2.B

35 seconds, 700 frames, snapshot 256 bytes; times in local time (the file stores UTC)



Fictional data. Cyber.SoHo / Asen Grozdanov, 420-1C1-DW Preventative Monitoring, Class 13.

Figure 4.2.2: the sample capture read by Code 4.2.B

Code example 4.2.A: the capture plan and the capture record (bash)

- Settings = the plan as variables: authorization, host, interface, filter, snapshot length, rotation (3600 s), number of files (8), folder, file pattern, record, minimum free space
- Check 1: the interface under /sys/class/net; Check 2: tcpdump with command -v and its version; Check 3: free space with df -m
- Builds: tcpdump -i enp0s8 -n -s 256 -G 3600 -W 8 -w ./captures/pc-acct-01_%Y-%m-%d_%H%M.pcap -Z tech 'host 10.20.1.101 and not port 514'
- Starts tcpdump only with --run, tcpdump present and root; otherwise 'Dry run'; the start time is taken in UTC
- Appends the record: start, operator, mode, host, interface, filter, snapshot length, rotation, files, pattern, tool, authorization, first file, SHA-256
- Run: bash class13_capture_plan.sh (dry run); sudo bash class13_capture_plan.sh --run on your MON-01

Code example 4.2.B: conversations, names and rate from a capture file (Python)

- Reads the 24-byte global header: magic number (little-endian, microseconds), version 2.4, snapshot length 256, link type 1 (Ethernet)
- Walks the records: 16-byte record header (time stamp, bytes kept, bytes on the wire); 453 of the 700 frames were cut by the snapshot length
- Decodes Ethernet (ARP counted, IPv4 decoded), IPv4 (header length, protocol, addresses), TCP (ports, sequence numbers, flags) or UDP (ports)
- Three detectors: the handshake tracker (SYN, SYN-ACK, ACK), the TLS server name of a ClientHello, the DNS answers (name behind each address)
- Conversation table with a fixed order of the two endpoints; bytes per second sent by the target; projection over one hour
- Prints the report and writes class13_conversations_output.csv; run: `python3 class13_pcap_conversations.py [file.pcap]`

Real-life example 4.2: deliverables and the lab they feed

- Deliverable 1: the capture plan and the capture record (captures/capture_record.csv) written by Code 4.2.A: the method section of the Lab 3 report
- Deliverable 2: the conversation table (class13_conversations_output.csv) with the DNS names, the handshakes and the upload rate printed by Code 4.2.B: the evidence section
- What goes to the diagnosis section: the interpretation, with the limits stated (no program, no account in the capture)
- In Lab 3: run 4.2.A with --run on your MON-01 during a replayed night, then 4.2.B on your own file; open the file in Wireshark with the filters of Table 4.2.C

Flash cards and self-check 4.2

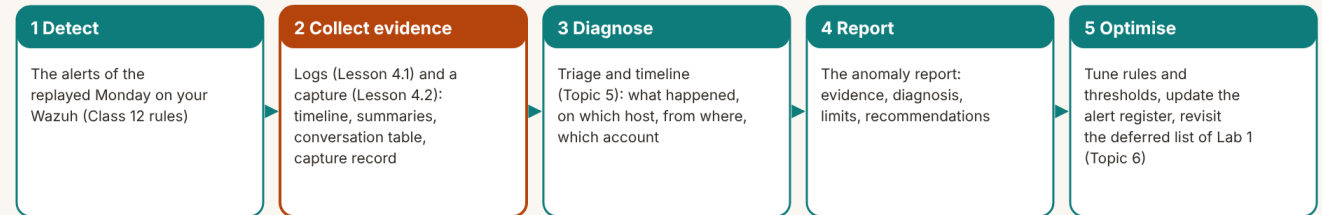
- Five flash cards: encapsulation and the four layers; capture filter versus display filter; the snapshot length; the capture points at the company and in the lab; what stays visible when the content is encrypted
- Self-check: four questions with one answer, two with several answers (select 3 and select 2)
- Every option has its own explanation; Hint helps without giving the answer
- On the lecture page, under Lesson 4.2

Lab 3 (5 %): Integrative Project, Part 3, summary (Figure 4.L3)

- Released as a separate guide on Léa (MyDawson portal): the guide alone gives the deliverables, the due date and the grading grid
- 1 Detect: replay a Monday through your Lab 2 rules. 2 Collect: Windows exports (local and central), the MON-01 authentication log, a capture of your PC-ACCT-01 with its record
- 3 Diagnose: timeline, triage (Lesson 5.2, Class 15), what happened, where, from where, which account, with the limits. 4 Report: evidence, diagnosis, the seven checks, recommendations (Lesson 5.5)
- 5 Optimise: tune rules and thresholds, update the alert register (Table 3.5.G), revisit the deferred list of Lab 1 (Topic 6)
- The four examples of today are models, not answers: your numbers come from your own lab

Figure 4.L3 Lab 3 at a glance: detect, diagnose and report an anomaly, then optimise

Integrative Project, Part 3, on the Lab 2 installation (summary only; the guide on Léa gives the deliverables, the due date and the grid)



Today's deliverables feed stage 2:

- Evidence timeline and per-account summary (codes 4.1.A, 4.1.B)
- Capture plan and capture record (code 4.2.A)
- Conversation table, DNS names, upload rate (code 4.2.B)
- The seven checks of Table 4.1.H applied to your lab

Isolated lab only, synthetic data, no automated blocking: Legal Notice sections 6.4, 7.2 and 7.4. Cyber.SoHo / Asen Grozdanov, Class 13.

Figure 4.L3: Lab 3 at a glance

Rules that apply to Lab 3

- Work only in your Authorized Lab Environment and on the Designated Targets named in the guide (Legal Notice sections 6.1, 6.4)
- Capture only on your own lab machines, never in promiscuous mode on a shared network; every capture and log is personal information (sections 7.2, 6.3)
- Synthetic data only: no real account, address or organization (sections 6.7, 18)
- tcpdump, Wireshark and Python from their official sources only, checksums verified (section 6.6, Annex C)
- No automated blocking: alerts inform, people decide (section 7.4)
- Confidential material, no recording, lab data deleted after marking (sections 17, 7.2); Annex A signed before any lab access

Links: [Legal Notice v2.0](#)



What to remember

- A log entry answers six questions; the fields differ by source, the questions do not
- Windows: the Security channel, its event numbers (4624, 4625, 4672, 4720, 4732, 1102), logon types and SubStatus codes; the audit policy decides what exists; a local log can be cleared or overwritten
- Linux: syslog lines in /var/log and the journal read by journalctl; grep, awk, sort, uniq; know the message formats
- Central collection with a common schema and 90 days of retention; seven checks before trusting a log
- A capture is a copy taken at a capture point the host does not control, with a filter, a snapshot length, bounded files and a capture record; targeted, not permanent
- Read a capture through its summaries; encrypted content stays hidden; the program and the account come from the host logs
- Every log and capture is personal information; capturing a shared network is interception

Next steps

- Quiz 3 (5 %): announced on Léa with its date; covers all the material to date, including today; prepared separately
- Class 14, Friday, October 9: Topic 4 continues with Lessons 4.3 and 4.4, the tools that search and filter the collected data, with today's exports and capture as material
- Lab 3: read the guide on Léa before Class 14; sign Annex A if not done; no lab access without it
- Tonight: the flash cards and self-checks of both lessons; the two videos of 4.1 and the two of 4.2
- Files to keep: the four programs, the sample capture, the two exports, the log, the tables and the workbook (companion ZIP)

Links (1/2): course, law, standards

- Cyber.SoHo Legal Notice v2.0: <https://cyber-soho.github.io/prevmon/00.01.%20cyber.soho%20-%20legal.html>
- Course page: <https://cyber-soho.github.io/prevmon/>
- Criminal Code, s. 184: <https://laws-lois.justice.gc.ca/eng/acts/C-46/section-184.html>
- Québec P-39.1 (Law 25): <https://www.legisquebec.gouv.qc.ca/en/document/cs/P-39.1>
- RFC 5424 syslog: <https://www.rfc-editor.org/rfc/rfc5424> RFC 5905 NTP: <https://www.rfc-editor.org/rfc/rfc5905>
- NIST SP 800-92: <https://csrc.nist.gov/pubs/sp/800/92/final>
- CIS Control 8: <https://www.cisecurity.org/controls/audit-log-management>
- CCCS ITSAP.80.085: <https://www.cyber.gc.ca/en/guidance/network-security-logging-monitoring-itsap80085>
- Check every link on VirusTotal first: <https://www.virustotal.com/gui/home/url>

Links: [Legal Notice](#) [Course page](#) [Criminal Code s. 184](#) [P-39.1](#) [RFC 5424](#) [NIST SP 800-92](#) [CIS Control 8](#) [CCCS ITSAP.80.085](#)



Links (2/2): documentation and tools, and the legal line

- Microsoft: event 4625 <https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-4625>
- Microsoft: logon types <https://learn.microsoft.com/en-us/windows-server/identity/securing-privileged-access/reference-tools-logon-types>
- Microsoft: wevtutil <https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/wevtutil>
- journalctl: <https://www.freedesktop.org/software/systemd/man/latest/journalctl.html> rsyslog: <https://www.rsyslog.com/>
- tcpdump: <https://www.tcpdump.org/> pcap-filter(7): <https://www.tcpdump.org/manpages/pcap-filter.7.html>
- Wireshark: <https://www.wireshark.org/> display filters: <https://www.wireshark.org/docs/dfref/>
- VirtualBox networking: <https://www.virtualbox.org/manual/ch06.html> Wazuh localfile: <https://documentation.wazuh.com/current/user-manual/reference/ossec-conf/localfile.html>
- Educational material, Cyber.SoHo / Asen Grozdanov, © 2026, all rights reserved; isolated authorized lab only; Criminal Code ss. 342.1, 342.2, 430(1.1), 184; Legal Notice v2.0, section 7

Links: [Event 4625](#) [Logon types](#) [journalctl](#) [tcpdump](#) [Wireshark](#) [VirtualBox ch. 6](#) [Wazuh localfile](#)

